Optimized Index Tracking using a Hybrid Genetic Algorithm

Roland Jeurissen and Jan van den Berg
rjeurissen@gmail.com j.vandenberg@tudelft.nl

***Abstract*— Assuming the market is efficient, an obvious portfolio management strategy is passive where the challenge is to track a certain benchmark like a stock index such that equal returns and risks are achieved. A tracking portfolio consists of a (usually small) weighted subset of stock funds. The weights are supposed to be positive here which means that short selling is not allowed. We investigate an approach for tracking the Dutch AEX index where an optimal tracking portfolio is determined. The optimal weights of a portfolio are found by minimizing the tracking error for a set of historical returns and covariances. The overall optimal portfolio is found using a hybrid genetic algorithm where each chromosome represents a specific subset of the stocks from the index, the fitness function of each chromosome corresponds to the minimized tracking error achievable with that subset, and the optimal portfolio is the tracking portfolio with highest fitness achievable. We show the experimental setup and the simulation results, including the out-of-sample performance of the optimal tracking portfolio. The hybrid genetic algorithms used appear to be robust in finding the optimal tracking portfolio and the performance of this portfolio on the out-of-sample data set is approximately four times better than that of randomly selected portfolios with optimized stock weights. By choosing a dedicated crossover operator, the hybrid genetic algorithm appears to find the optimal tracking portfolio using, on average, less than 23 generations only.**

***key words*: index tracking, hybrid genetic algorithm**

I. Introduction

A. *Portfolio management strategies*

Portfolio management approaches can be divided in three categories [2] namely *active management*, *passive management*, or a *mixed strategy* of the two. Active management relies on the belief that skillful investors are able to outperform the aggregate market return. An active manager can try to compose dedicated portfolios that outperform other collections of stocks or may try to exploit arbitrage opportunities like market timing where the moments of buy and sell decisions are optimized. These opportunities are supposed to exist based on the belief that financial markets are not fully efficient. Passive management on the other hand is adopted by investors who believe that financial markets are efficient. They believe that it is impossible to consistently beat the market. Their main activity is endeavoring to achieve the same returns and risk of a certain benchmark like a well-diversified portfolio like a stock index. Attempting to reproduce the performance of an index, such as the S & P 500, is often referred as *index tracking*. A passively managed fund whose objective is index tracking is called an *indexed exchange-traded fund* (ETF) [3] or, shortly, a *tracker fund*.

The debate about active versus passive management is still continuing. Sharpe [4] asserts that on average active managers can not beat passive strategies. Active investing is a zero sum game where some investors will win and some will lose relative to the return of the market or a market segment. Consequently, after costs, the return on the average actively managed dollar will be less than the return on the average passively managed dollar. Empirical researches by Malkiel [5], Sorenson [6] and Frino [7] have shown that passive strategies are found to outperform active strategies on average. So it is not a surprise that passive strategies are increasingly popular. Frino [7] estimates that in total $1,5 trillion is passively invested in index funds in the USA alone.

Matching the performance of an index can be performed in two ways. The first is *full (or complete) replication*, a method in which all the shares making up an index are held in their respective market weights. This method perfectly reproduces the index, but incurs high transaction costs. E.g., imagine buying all the stocks of the Wilshire 5000 composed of more than 6000 companies. Moreover, when stock weights comprising the index change (as a result of a merger, stock split etc.), high transaction costs will occur again. Unfortunately, composition change of indices happens a lot. The process of adjusting a tracking portfolio due to changes of an index is referred as *rebalancing*.

For the above-given reasons, many index fund managers hold only a subset of stocks from the index. This method is called *partial replication* and involves lower transaction and management costs. However, this strategy introduces a *tracking error*, the measure of the deviation of the chosen portfolio from the index [8]. Generally, performance deviations of the tracking portfolio are supposed to arise through several factors. The main factors identified by Frino and Galagher [7] are transaction costs, the way dividends are treated by the index, the volatility of the index, timing effects, front-running by arbitrageurs and index composition changes. They also noticed seasonality of the tracking error: the tracking error is significantly higher in January and during months when stocks go ex-dividend. Liquid indices such as the S & P 500 tend to have low transaction costs compared to non-liquid ones, but investors have to pay a substantial price premium for stocks in these markets as identified by Petajisto [9]. When a new stock is added to the index, this generally goes up. This results into extra costs for index managers because they have to buy the stock at the new price. Larsen-Jr. and Resnick [10] have shown that tracking portfolios for value weighted indices (based on market capitalization) have a less

Roland Jeurissen is with BNP Paribas and Jan van den Berg with Delft University of Technology.

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tracking error and lower standard deviations of the tracking error than they have for equal weighted indices.

Obviously, index fund managers try to minimize the tracking error. It should further be clear from the above-given discussion that the index tracking problem actually consists of two subproblems, namely, (*i*) the minimization of the tracking error and (*ii*) the minimization of transaction costs.

B. Literature review

Looking at *traditional methods* of index tracking, we have noticed that there exist several models for reproducing the performance of an index using only a subset of stocks. Meade and Beasley [11] and Larsen-Jr. and Resnick [10] used a single factor model in order to minimize tracking error given a subset of shares. Meade and Salkin [12], [13] discuss some assumptions related to the tracking error in order to solve the problem by using quadratic programming. They also considered the effect of industry stratification within a tracking portfolio. However, this strategy did not boost the performance in their research. On the other hand, Larsen-Jr. and Resnick [10] found out that stratification does work, especially for high capitalization indices.

Tabata and Takeda [14] considered the traditional asset allocation of the Markowitz type [15], [16] and used an efficient approach to find a local optimal solution. They also pointed out that the problem is a bi-criteria optimization problem: firstly the zero-one integer problem deciding which stocks to include in the portfolio must be solved and afterwards an algorithm solving the weights of the chosen stocks by minimizing tracking error must be applied. However, this problem has no tractable analytic solution. To see why consider the problem of tracking the S & P 500 with 200 stocks. The number of possible portfolios equals $500!/(200! * 300!) \approx 5,05 * 10^{144}$. So, the solution space is very large because of its combinatorial explosion. That is why many researchers proposed *heuristic approaches* for this problem, either of stochastic or of deterministic nature.

Gilli and Kllezi [17] used a *Threshold Accepting Algorithm* which is a refined local search procedure which escapes local minima by accepting solutions which are not worse by more than a given threshold. The algorithm is deterministic as it does not depend on some probability. The constructed portfolios performed rather well for different data sets. In their objective function they also considered transaction costs. Another heuristic search approach is *simulated annealing* [18]. This procedure is implemented for the tracking error problem by Derichs and Nickel [19]. Besides the creation of a successful tracking portfolio, they also considered the revision - including the transaction costs involved - of an existing tracking portfolio over time. Next, Dose and Cincotti [20] formed a tracking portfolio based on *complete-link hierarchical clustering*. They cluster homogeneous groups based on similarity of returns and then take one stock from each cluster to form a subset. After the selection they perform weight optimization of the subset. Their results are not as good as found by the previous methods. Last but not least is the use of *genetic algorithms* in order to search the solution space for a subset of shares that performs well in tracking an index. When this subset is determined, optimization techniques such as quadratic programming are used to solve the optimal proportions to hold the shares. Shapcott [8] is, to our knowledge, the first who successfully employed this strategy. He tracked the FTSE 100 index (UK) effectively with a portfolio consisting of 20 stocks. He also proved the superiority of the algorithm compared to random search algorithms. Eddelbüter [21] used a similar strategy when tracking the German Xetra DAX 30 index. The research included also a section describing the fact that the genetic algorithm was able to converge to the global minimum tracking error portfolio. Beasley et al. finally [2] extended these approaches by considering transaction costs and the revision of an existing tracking portfolio. They tested the approach on the S & P 500 (USA), the FTSE 100 (UK), Hang Seng (Hong Kong), Xetra DAX 100 (Germany) and the Nikkei 225 (Japan). The results are quite impressing. They also conducted reduction tests in order to reduce the size of the search space and hence enabling the algorithm to be more effective. Because of the completeness of their research, this is the leading paper for index tracking using genetic algorithms.

C. Goal of this paper

Inspired by the outstanding results of the genetic algorithm, we decided to use this approach for tracking the Dutch AEX index. We will assume that the number of stocks to be selected is fixed and that the composition of the chosen tracking portfolio is not changed which implies that we will not analyze optimal timing strategies and, related to that, not take transaction costs into account. Instead, we will focus, in two ways, on the minimization of the tracking error by determining (a) the optimal subset of index stocks in combination with (b) the optimal weights of these stocks in the tracking portfolio. The findings of this type of research are presented in this paper. The rest of this paper is structured as follows. In the next section II, we sketch our approach of tackling the index tracking problem, in section III, we discuss the main results obtained and in section IV we make some conclusions and present an outlook.

II. Research Approach

A. Formulating the problem

Let $r_B = \mathbf{b}'\mathbf{r}$ represent the return of the benchmark (index) portfolio B consisting of N stocks where $\mathbf{b}' = (b_1, \ldots, b_i, \ldots, b_N)$ is the transposed vector of $\mathbf{b}$ defining the weights b_i of the stocks i in this benchmark portfolio and $\mathbf{r}$ is the vector of returns. Each weight b_i represents the percentage of the market capitalization (aggregate value) of the stocks i in the benchmark portfolio B (which, for the sake of clearness, is generally different from the relative number of stocks i present in the benchmark portfolio). The variance σ_B^2 of portfolio B can be expressed as $\sigma_B^2 = \mathbf{b}'\mathbf{V}\mathbf{b}$ where $\mathbf{V}$ is the symmetric matrix of covariances [21].

Partial replication involves the inclusion of a subset of stocks in the passive portfolio that tracks the benchmark. As has been mentioned above, we shall assume that the number of stocks in this subset is fixed and given beforehand. Let $r_P = \mathbf{t}'\mathbf{r}$ represent the return of the tracking portfolio P where $\mathbf{t}$ is the vector defining the weight values t_i of the stocks in the tracking portfolio. The variance of portfolio P can be expressed as $\sigma_P^2 = \mathbf{t}'\mathbf{V}\mathbf{t}$. We further assume that for both portfolio's the weight values of their stocks sum up to one, i.e., $\sum_i^N b_i = 1$ respectively $\sum_i^N t_i = 1$, and that these weight values are non-negative, i.e., short selling is not allowed.

The objective function to be minimized is the tracking error of the passive portfolio P with respect to its benchmark portfolio B. There exist several ways to define this tracking error [7]. In this paper, we will use the standard test used in industry called the 'volatility of the tracking error' defined by the expected variance of the difference between the returns of the passive portfolio and those of the benchmark. The standard deviation of this variance can be written as [21]

$$\sigma_{r_P - r_B} = \sqrt{(\mathbf{t}-\mathbf{b})'\mathbf{V}(\mathbf{t}-\mathbf{b})} = \sqrt{\mathbf{x}'\mathbf{V}\mathbf{x}}. \qquad (1)$$

Note that the tracking error defined by equation (1) focusses on minimizing the variance/standard deviation in the return differences. So the natural requirement (constraint) of selecting a passive portfolio having the same expected return as the benchmark is simply discarded [21].

Having chosen the above-given approach, the problem of finding an optimal tracking portfolio consists of two parts. The first part consists of selecting the right subset of stocks, the second part of minimizing equation (1) by changing the weights t_i of the selected stocks in the tracking portfolio. As weights b_i of the benchmark portfolio, we simply take their current weights in the index. Minimizing equation (1) involves a *quadratic programming* problem, i.e., it concerns the minimization of a quadratic function subject to linear constraints. More precisely, we can formulate this quadratic programming problem as: given $\mathbf{V}$ and $\mathbf{b}$,

$$\begin{gathered} \text{minimize } \sqrt{\mathbf{x}'\mathbf{V}\mathbf{x}} \\ \text{subject to } \mathbf{1}' \cdot \mathbf{x} = 0 \text{ and } -\mathbf{b} \le \mathbf{x} \le \mathbf{1} - \mathbf{b}. \end{gathered} \qquad (2)$$

The constraints in (2) are derived as follows. Since the weights b_i of the index and the weights t_i of the tracking portfolio portfolio sum to 1, it follows that the sum of their differences $x_i = t_i - b_i$ must be equal to 0 or, shortly, $\mathbf{1}' \cdot \mathbf{x} = 0$ where $\mathbf{1}$ is a vector containing only ones. The inequality in (2) follows directly from the fact that $\mathbf{0} \le \mathbf{t} \le \mathbf{1}$ by subtracting $\mathbf{b}$ from all the terms.

B. Finding the optimal portfolio

A so-called *hybrid genetic algorithm* [21] is used to solve the above-formulated problem of finding the optimal tracking portfolio. A genetic algorithm is used to search the solution space for possible subsets of stocks. The genetic algorithm (GA) is a randomized search algorithm based on mechanics of natural selection and genetics [22]. The evolution starts from a population of randomly chosen individuals represented by chromosomes. Here, in each generation, the fitness of all population members is determined using the solution of the quadratic programming problem (2). Multiple individuals are selected from the current population based on their fitness value found. Next, the selected individuals are modified using 'crossover' and 'mutation' operators to form a new population [23]. Each chromosome represents a possible solution, here, a subset of shares from the benchmark portfolio. Typically, binary strings are used as an encoding. Since the GA-approach is combined with a quadratic programming approach, the GA is termed 'hybrid'. Some authors (including those of [24]) prefer to use the name 'memetic' algorithm in case there is a 'marriage between a population based global search and a heuristic local search made by each of the individuals'. This quote is taken from [25] where the term 'memetic algorithm' is used for the first time. In this paper however we use the name hybrid genetic algorithm inspired by the notion of hybrid scheme 'that exploits the global perspective of the GA and the convergence of a problem specific technique' [26].

C. Experimental setup

In this subsection we present a lot of the implementation details related to the simulations we performed.

1) Data set: The AEX-index is the best known index of Euronext Amsterdam and is made up of 25 high capitalized stocks. The weights of the stocks are based on market capitalization and are adjusted real time. The index provides a fair representation of the Dutch economy (more details about this index are available at http://www.euronext.com). Our goal is to reproduce the performance of this index using only a subset consisting of 10 index shares.

The composition of the AEX-index appears to change often. For example, during the time between 01/01/2001-04/05/2004, the index has changed 17 times. We used the data between 02/01/2001-02/03/2004 as *training data* for finding the optimal tracking portfolio. We tested the tracking portfolio found *out of sample* using the data from the period 03/03/2004-03/03/2005. We used the composition of the index on 02/03/2004 to determine the benchmark weights (for precise information on the benchmark weights used, we again refer to [1]). The covariance matrix is calculated using daily stock quotes between 02/01/2001-02/03/2004 (calculation details are given below). All quotes are obtained by using DATASTREAM (http://www.datastream.net/) and are adjusted for stock splits, dividends etc.

2) Implementation of genetic operators: Since we need to select 10 stocks from an index having 25 stocks, each chromosome can be represented by a binary string consisting of 25 bits, 10 of which are equal to '1'. The fitness function evaluates the strength of each chromosome. A more fit individual has a higher probability of reproduction over a less fit one. In our research, the fitness function equals the calculated tracking error of each chromosome. The lower the better, because our objective is to minimize the tracking error defined by equation (1). In order to calculate the fitness of each chromosome, we do need to determine the optimal weight vector $\mathbf{t}$ of each chromosome by, first, solving minimization problem (2) and, second, based on the vector $\mathbf{x}$ found, calculating $\mathbf{t}$. At this stage, the genetic algorithm is hybridized with the quadratic programming routine. The routine solves the problem for each chromosome and delivers the corresponding tracking error to the fitness function of the genetic algorithm. This approach combines the search procedure of the genetic algorithm in the solution space with the local convergence properties of the quadratic programming solver.

In this paper we use deterministic tournament selection as *selection* operator. Tournament selection runs a tournament among a few individuals and selects the winner (the one with the best fitness). Tournament selection has several benefits: it is efficient to code, works on parallel architectures, and allows the selection pressure to be easily adjusted [27]. Associated with the selection step is the 'elitism' strategy. Elitism is a method that guarantees that a number of best solutions are placed directly into the next generation. In the approach presented in this paper, elitism is used, because that way, the search for a good solution never goes backwards.

Crossover operators (which take two parent chromosomes and combine them to produce a child) need to be carefully designed in order to guarantee that the number of ones and zero's (i.e., the number of stocks) in the children chromosomes does not change. To deal with this constraint, we tried two different crossover operators. First, a two-point 'order-based' crossover [28] has been used. The idea behind the order-based crossover is to swap the genes in the order found at the other parent, going from left to right. The number of ones and zeros will automatically remain the same. In

Parent 1	100**0011**011
Parent 2	011**0110**010
Cut Point	3 and 7
Child 1	100**0110**011
Child 2	011**1001**010

TABLE I

EXAMPLE OF A TWO-POINT ORDER-BASED CROSSOVER. THE BOLD PART IS SWAPPED IN THE ORDER FOUND AT THE OTHER PARENT, GOING FROM LEFT TO RIGHT. THE UNDERLINED PART REMAINS THE SAME.

this way, the substrings of the chromosomes on the left and righthand site are preserved while the bit substring in the middle part changes: for an example, we refer to Table 1. Unfortunately for this problem, order-based crossover yields a rather random change of bits in the middle part, i.e., it intrinsically involves a quite high mutation rate in the middle part of the chromosomes.

Therefore, we also tried another two-point crossover operator. This operator works as follows. If, after having generated two children using standard two-point crossover, the child chromosomes happen to contain the same number of zeros and ones as the parents, the crossover operation is finished. If the number of ones (respectively zeros) in a child chromosome is larger than that of the parents, then bits with value one (respectively zero) are randomly selected and changed into a zero (respectively one) until the number of zeros and ones equals that of the parents. In this way it is tried (*i*) to minimize the occurrence of randomness (mutation) by the crossover operator and (*ii*) to spread these (additional) mutations more uniformly. We term this second crossover operator the two-point 'bit equalizer' crossover operator.

Mutation is necessary to prevent areas of the search space being discarded. The standard mutation operator chooses a single bit at random and swaps its value. Again, we did not allow changing the number of ones and zero's. Therefore, we applied 'mutation inversion': rather than selecting a single bit to mutate, inversion mutation finds just two random characters in the string and reverses them. We further note that the mutation effect of this inverse mutation operator is uniformly spread over the chromosome and, in addition, generally lower that the mutation effect caused by the order-based crossover operator mentioned above since the latter often causes mutations of several bits.

3) Parameters of the Genetic Algorithm: One of the difficulties of genetic algorithms is finding the best internal parameters in order to optimize speed and convergence. These parameters are

- Size of the initial population
- Number of generations
- Crossover probability
- Mutation probability
- Rate of elitism.

The parameter values to be used are dependent on the type of GA operators used. In order to be able to compare the performance of the two crossover operators used, we tried to keep the rest of the parameters the same. After quite a lot of trial and error, the parameter values as presented in Table 2 appeared to be adequate. A rather low crossover probability (0.1) together with a high mutation probability (1.0) yielded the best results in the first series of experiments. This is remarkable since it actually means that the two-point order-based crossover does not really works fine. The fact that this way, eventually, optimal solutions are found yet seems to be based on an almost less random search (caused by the high mutation rate) in combination with the application of an elitism strategy.

This was the reason to look for another cross-over operator, the above-mentioned two-point bit equalizer crossover. In the second series of experiments while applying this operator, a more usual crossover probability (1.0) together with a more normal, lower (but still quite high) mutation probability (0.5) appeared to be best choices.

Initially, we used 50 generations of both GAs. For the GA with the two-point bit equalizer crossover operator, this number appeared to be large enough to guarantee convergence to the same final solution in all experiments. However, this number is not sufficient to guarantee convergence to the final solution in case of using the GA with the two-point order-based crossover operator. We therefore also tried that GA using 100 generations.

	(A)	(B)
Size of the initial population:	50	50
Number of generations:	50, 100	50
Crossover probability:	0.1	1.0
Mutation probability:	1.0	0.5
Rate of elitism:	3	3

TABLE II

INTERNAL PARAMETERS OF THE GENETIC ALGORITHM USING THE (A) TWO-POINT ORDER-BASED CROSSOVER, AND (B) TWO-POINT BIT EQUALIZER CROSSOVER OPERATOR.

4) Estimating the covariance matrix: In order to calculate the covariance matrix, we used equal weights for past returns. Returns have been calculated 'continuously' conform

$$r_n(t) = ln\Big(\frac{p_n(t)}{p_n(t-1)}\Big), \tag{3}$$

where $p_n(t)$ equals the price of stock n at time t. The covariance matrix $\mathbf{V}$ has been estimated by calculating the elements of matrix (r_n, r_m) according to

$$cov(r_n, r_m) = \frac{\sum_{t=0}^{T-1}(r_n(T-t) - \overline{r}_n)(r_m(T-t) - \overline{r}_m)}{T-1}, \tag{4}$$

where $\overline{r}_n$ is the arithmetic mean of historical returns $r_n(t)$.

5) Validation: As mentioned above, we applied out of sample testing in order to evaluate the performance of the optimal tracking portfolio. To do so, we calculated two tracking errors based on the data samples from the period 03/03/2004-03/03/2005. The first tracking error E_1 is directly related to the volatility of the tracking error and is based on an estimation of standard deviation (1). It is written as

$$E_1 = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}((r_P(t) - \overline{r}_P) - (r_B(t) - \overline{r}_B))^2}, \tag{5}$$

where $\overline{r}_P$ and $\overline{r}_B$ are the arithmetic mean of the returns of tracking portfolio and the benchmark portfolio respectively.

The second out-of-sample tracking error E_2 is defined as the average of absolute differences in the returns $r_P(t)$ of the index portfolio and the returns $r_B(t)$ of the benchmark. According to [29], E_2 can be stated as.

$$E_2 = \frac{\sum_{t=1}^{T}\sqrt{(r_P(t) - r_B(t))^2}}{T}. \tag{6}$$

6) Test environment: The returns and covariance matrix are calculated using Microsoft Excel 2003. This program is also used to evaluate the performance of constructed portfolios and to make the graphs that are presented in the results section. MATLAB 6.5 is used to code the genetic algorithm and it performs the quadratic programming. All computations were performed on an AMD Athlon$^{\text{TM}}$ 64 processor, 1.80 Ghz, 1 GB Ram, personal computer running Windows XP Professional.

III. RESULTS

In this section, we present our most important findings. We start by presenting the results related to the optimality of the solution found, next we present a performance analysis with respect to the optimal solution found, and finally we give some results concerning the applied GAs.

A. Optimality of solution found

We did several experiments and while using the above-given parameters, the hybrid genetic algorithm was able to find the solution presented in Table 3, no matter which of the two crossover operators was applied. Both the company names and their weight value in the optimal tracking portfolio are given. We observe that, according to what may be expected, the stocks selected for our optimal tracking portfolio coincide with the larger stocks of the AEX-index. In addition we note that the minimum weight in the tracking portfolio (that of AHOLD shares) is around 4,7% and the maximum weight (that of Unilever stocks) somewhat less than 14,5%.

Company	Weight
ABN AMRO	0,126271
Aegon NV	0,075173
Ahold	0,046730
Akzo Nobel	0,078453
Fortis	0,087774
ING	0,084059
KPN	0,075942
Philips	0,141958
Royal Dutch	0,139015
Unilever	0,144623
SUM	1

TABLE III

THE PORTFOLIO CONSTRUCTED BY THE GENETIC ALGORITHM.

Since the AEX-index consists of 25 stocks and is replicated by using just 10 stocks, we have also been able to check whether the GA tracker found is indeed the optimal portfolio with respect to the training data. We tried all possible combinations of 10 stocks out of 25 and calculated, based on the training data, the corresponding minimum expected tracking error as given by (2). The best solution found appeared to coincide with the solution found by hybrid GA. So, an important first conclusion of our experiments is that the hybrid GA is able to consistently find the optimal solution.

B. Quality Analysis of Solution Found

In order to estimate the quality of the GA tracking portfolio found, we performed several experiments. First of all, we measured the above-described out-of-sample performance. Figure 1 shows the out-of-sample performance compared to the performance of the true AEX-index. We observe that initially, both performances are almost equal while gradually small differences emerge due to, among other things, changes in the weights of the shares composing the AEX-index. We further note that the up and down movements of both portfolios are quite similar.

Fig. 1. Tracking performance of a portfolio (10 stocks) constructed by the Hybrid Genetic Algorithm.

Second, we calculated the out-of-sample tracking error as defined by equation (5), respectively equation (6). The error values found are presented in the first column of Table 4.

TE	GA Tr.	Rand. Av.	High Cap.	Low Cap.	Full Repl.
eq. (5)	.001170	.004373	.001611	.005449	.000122
eq. (6)	.001534	.004422	.001905	.006034	.000089

TABLE IV

OUT OF SAMPLE TRACKING ERRORS (TES) OF DIFFERENT PORTFOLIOS. ROW 1 SHOWS TRACKING ERRORS DEFINED BY EQUATION (5), ROW 2 TRACKING ERRORS DEFINED BY EQUATION (6). THE SECOND COLUMN (GA TR.) PRESENTS THE TRACKING ERRORS OF THE OPTIMAL GA TRACKER, THE THIRD COLUMN (RAND. AV.) THE AVERAGE TRACKING ERRORS OF 50 RANDOMLY SELECTED TRACKERS, THE FOURTH COLUMN (HIGH CAP.) THE TRACKING ERRORS OF THE TRACKER CONSISTING OF THE 10 HIGHEST CAPITALIZED STOCKS, THE FIFTH COLUMN (LOW CAP.) THE TRACKING ERRORS OF THE TRACKER CONSISTING OF THE 10 LOWEST CAPITALIZED STOCKS, AND THE LAST COLUMN (FULL REPL.) THE TRACKING ERRORS OF THE PORTFOLIO REPLICATING THE FULL INDEX.

In trying to understand the quality of these error values, we compared them with the performance of randomly selected portfolios. To do so, 50 portfolios consisting of 10 randomly selected stocks were constructed and the optimal weights of these stocks were calculated by solving minimization problem (1). Next, we calculated the out-of-sample performance of the randomly selected portfolios according to equations (5) and (6), the averages of which are shown in the second column of table 4. We conclude that, roughly spoken, the optimal GA tracker, on average, performs almost four times better than a randomly selected portfolio with optimized stock weights.

We further selected the 10 highest, respectively lowest capitalized stocks in the AEX-index using equal weights, the tracking errors of which are shown in the fourth column of table 4 and the tracking behaviors of which are shown in figure 2. Actually, the tracking errors of the portfolio consisting of the 10 highest capitalized stocks appear to be quite small although the performance in terms of the value of this portfolio is slightly worse than that of the index. In addition we observe that the tracking errors of the lowest capitalized stocks in the AEX-index appear to be higher while the performance in terms of the value of this stock is much better than the AEX-index. Finally we applied full replication of the AEX-index, the performances of which are shown in the last column. As expected, these performances are very well although not 100% due to (small) intermediate changes of the composition of the AEX-index in the out-of-sample test data.

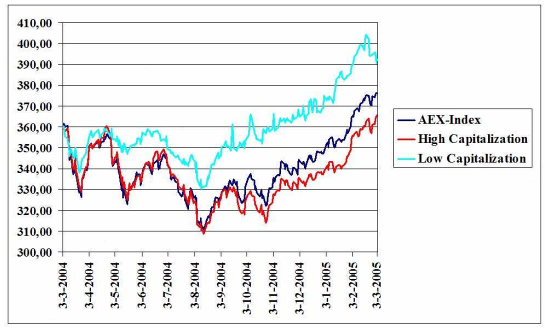

Fig. 2. Tracking performance of respectively the highest and lowest capitalized portfolios, in both cases consisting of 10 stocks.

C. Performance Analysis of the Hybrid Genetic Algorithm

In this subsection we mention some results related to the performance of the hybrid GA, especially with respect to the crossover operators used. When comparing the performance of the two cross-over operators, we initially used equal values for the other parameters like crossover and mutation probabilities as well as the elitism rate. It turned out that in those cases, the bit equalizer crossover operator worked generally much better.

Next we performed experiments with the optimized parameter values as mentioned in table II. As was already mentioned above, both the GA with the two-point order-based crossover and the GA with the two-point bit equalizer crossover operator were able to find the optimal solution. In all cases, we tried the GAs a hundred times using different initializations. In case of using the GA with the two-point order-based crossover with 50 generations (and other parameter settings as mentioned in Table 2), the optimal solution was not always found, namely in just 89% of the cases. However, in case of using that GA with 100 generations, the optimal solution was always found in, on average, almost 31 iterations with a standard deviation of approximately 15,5.

In case of using the the GA with the two-point bit equalizer crossover operator a hundred times with 50 generations, the GA was able to find the optimal solution in all cases using, on average, only 22,2 iterations with a standard deviation of approximately 9,8 iterations. These numbers show that the GA with the bit equalizer crossover operator is both robust in finding the optimal solution and fast, although the speed of convergence is somewhat sensitive for the chosen initial population. The following two figures further illuminate these findings. Both the progress of the average fitness value

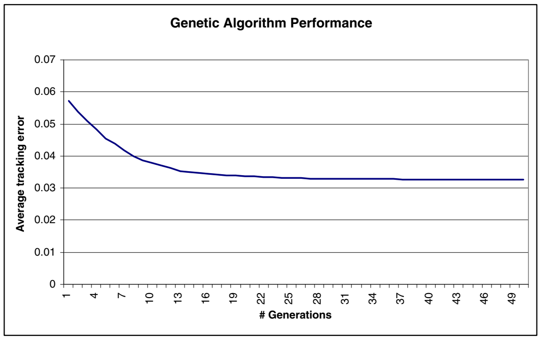

Fig. 3. Progress of the average tracking error of best solution in each generation using the hybrid GA with bit equalizer crossover operator.

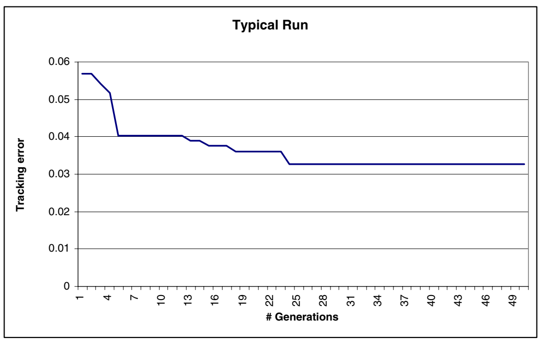

Fig. 4. Progress of tracking error of best solution in each generation of a typical run using the hybrid GA with bit equalizer crossover operator.

(tracking error) of the best solution of each generation is shown and the progress of the fitness value of the best solution of each generation in a single, typical run are shown. For certain other results found concerning the AEX-index tracking problem, we refer to [1].

IV. Conclusions and Outlook

In this paper, we presented a GA approach for finding an optimal tracking portfolio of the Dutch AEX stock index. The overall optimal portfolio is found using a hybrid (memetic) genetic algorithm where the fitness function of each chromosome (possible subset of stocks) equals the minimal tracking error achievable. This minimal tracking error is determined by solving the corresponding quadratic programming problem. We have shown the experimental setup of the experiments as well as the simulation results.

We conclude that the performance of the tracking portfolio found is much better than that of randomly selected portfolios and that of low capitalized portfolios which can be derived from the AEX-index. In addition, it is shown that the performance of high capitalized portfolios is similar although worse than the optimal GA tracking portfolio. This research shows that even a small stock index like the AEX-index can be tracked quite well by a small subset of its composing stocks.

We also analyzed two types of crossover operators. Generally spoken, the two-point order-based crossover performed much less than the two-point bit equalizer crossover operator. We further conclude that for our problem, that of tracking the AEX-index, the two-point bit equalizer crossover operator with optimized parameter settings always yields the optimal solution using a relatively small number of generations namely, on average, less than 23 generations.

Further research might show whether portfolios consisting of subsets with a number of stocks different from, but close to 10 yield similar results. Furthermore, it seems worthwhile to analyze whether the robust behavior of the hybrid GA with the two-point bit equalizer crossover operator also holds in cases of tracking stock indices consisting of hundred or more stocks. In addition to this, it seems worthwhile to look for alternative crossover operators that still better perform.

Finally it looks interesting to analyze the frequency needed for re-balancing the optimal tracking portfolio in an attempt to further improve the performance of the GA tracker in the long run. It is also interesting to investigate what the results will be in case short selling is allowed.

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